A32NX exterior: spoilers by panel group and visible RAT

The A32NX exterior spoilers move by panel group from their own deflection instead of the wing average, and the ram air turbine (RAT) is now visible on the exterior model

Aircraft	A32NX
Where to find it	Outside view of the wings and of the left belly fairing
Branch	`core/a32nx-exterior`, based on `master` (FlyByWire master 76db0aef1 merged in); independent of the other core branches. The exterior model and the new RAT model (fbw-a32nx/src/model/rat, generated by generate-rat.js) are in the main repository (no aircraft-large-files change). The RAT goes back in only after a ground stow, which is on `core/rat-ground-stow` (flyPad row on `core/efb-redesign`).
Status	Spoilers: tested in the sim by the user: OK. RAT model: tested in the sim by the user: OK (deploy and stow; with the fixed model the aircraft loads and the RAT deploys and turns).
Pictures	No pictures: the movements do not show well on a still picture.

1. Exterior spoilers move by panel group

> **Aircraft:** A32NX | **Where:** Outside view of the wings | **Status:** Tested in the sim by the user: OK.

What it is for

Each wing has five spoiler panels: 2 to 5 help the ailerons in roll, 2 to 4 are the speedbrakes, and all five are ground spoilers (A320 FCOM DSC-27). The exterior model moved all the panels of a wing together from the wing average, so spoiler 1 rose in turns and with the speedbrakes. The panels now follow their own deflection from the hydraulic model, by group, and the angle is shown in true degrees.

- Three groups per wing: panel 1, panels 2 + 3, panels 4 + 5. Panels 2 and 3 share their skin joints in the model, and so do panels 4 and 5, so a group shows the mean of its two panels.
- True degrees: the model animation goes to 45 degrees; the ground spoilers (50 degrees) show as the model maximum.

What you should see

Situation	What you should see
Roll in flight	The panels 2 to 5 of the down-going wing rise (as two groups); spoiler 1 stays down.
Speedbrakes extended	Panels 2 to 4 rise; spoiler 1 stays down; the 4 + 5 group shows the mean of panels 4 and 5 (half way when only panel 4 is out).
Ground spoilers at landing or rejected takeoff	All three groups rise to the model maximum.

Limitations and why

Limitation	Why	What to do
Panels 2 / 3 and 4 / 5 move together.	Their vertices are blended between two skin joints in the model; five separate panels need an edit of the model skin weights.	None for now.

A full MSFS restart is needed after an update.	The exterior model and its behaviour XML change.	Restart the sim after installing.

How to use

1. Nothing to set: fly as usual and watch the wings from the outside view or the cabin.

How to test

#	Do this	Expected result
1	A32NX in flight, outside view behind a wing. Roll left with the sidestick.	On the left wing, the spoiler groups 2 + 3 and 4 + 5 rise; spoiler 1 (inboard) stays down. The right wing spoilers stay down.
2	Level the wings and extend the speedbrakes fully.	Panels 2 to 4 rise on both wings; spoiler 1 stays down; the outboard 4 + 5 group is lower than the 2 + 3 group.
3	Retract the speedbrakes. Land with the ground spoilers armed.	At touchdown all spoilers of both wings, spoiler 1 included, rise fully.

2. Visible ram air turbine (RAT)

> **Aircraft:** A32NX | **Where:** Outside view, under the left belly fairing | **Status:** Tested in the sim by the user: OK (deploy and stow, with the fixed model the aircraft loads and the RAT deploys and turns, RAT flight R1). Unit tests: RatModel.spec.ts.

What it is for

The A32NX RAT was only a system: the HYD page showed it out, but nothing showed on the aircraft. A RAT model is now part of the exterior (the two most detailed levels of detail, LOD 0 and 1), driven by the hydraulic system values.

- Stowed: the fuselage skin closes the bay; the doors, the bay and the turbine are hidden.
- Deploying: the doors under the left belly fairing open over the first 20 % of the travel, then the strut swings down.
- The propeller (blades 0.749 m across, the real 29.5 in RAT) turns in the airflow at the system RAT speed. Until the RAT is 80 % out the blades stay lined up with the strut, so they pass through the slot.
- Above 500 rpm a blur disc replaces the blades (a turning blade would only strobe on screen).
- The blade pitch follows the RAT governor (fine at low speed, coarser when governed).
- After a ground stow the strut swings back in and the doors close.

What you should see

Situation	What you should see
RAT stowed	No RAT and no doors: a closed belly fairing.
RAT extending (both AC buses lost, RAT MAN ON or EMER ELEC PWR MAN ON)	The doors open, the strut swings down, the propeller starts to turn once the RAT is almost fully out.
RAT out in flight	A blur disc at the propeller.
RAT stowed on the ground (flyPad Ground > Services > RAT stow)	The strut swings in and the bay closes.

Limitations and why

Limitation	Why	What to do

The RAT goes back in only after a ground stow.	The stow request is on `core/rat-ground-stow` (systems) and `core/efb-redesign` (flyPad row); without them the RAT stays out for the rest of the session, as on master.	Merge both branches.
Only LOD 0 and 1 have the RAT.	It is small and under the aircraft: the farther levels of detail leave it out.	-
Position and size were measured on another A320 model.	No Airbus drawing was available.	-
A full MSFS restart is needed after an update.	The exterior model, models.json and the behaviour XML change.	Restart the sim after installing.
The first version made the whole A32NX invisible.	MSFS refuses an animation buffer view that mixes time and rotation data. The generator now gives each animation its own buffer view, and RatModel.spec.ts checks it.	-

How to use

1. Nothing to set: when the RAT extends, look at the left belly fairing from the outside view.
2. After the landing, stow it from flyPad > Ground > Services (needs `core/rat-ground-stow` and `core/efb-redesign`).

How to test

#	Do this	Expected result
1	Load the A32NX after a full MSFS restart. Outside view.	The aircraft is visible; no RAT and no doors under the belly.
2	In flight, about 5,000 ft, 200 kt. Overhead HYD panel: open the guard and press RAT MAN ON. Outside view of the left belly fairing.	The doors open, the strut swings down, then the propeller turns and shows as a blur disc. The ECAM HYD page shows the RAT out.
3	Land and stop. flyPad > Ground > Services: press Stow on "RAT stow (maintenance)" (needs `core/rat-ground-stow` and `core/efb-redesign`).	The strut swings back in and the bay closes.